Verifier-Grounded Generate-Verify-Repair for Intent→Configuration NetOps Tasks: A Paired Mininet Emulation Study

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***Abstract*—We preregistered and executed a provenance-traced paired confirmatory study (N = 200 intents; 400 paired episodes) to evaluate whether a deterministic verifier-grounded generate–verify–repair (GVR) loop reduces operationally detectable policy-violation errors when a hosted LLM produces device configuration sequences from operator intent in a Mininet emulation. Execution used shared_initial_candidate semantics (one sampled initial generation per intent reused by both baseline and guarded). Baseline success (no detected policy violation) = 196/200 (98%); guarded success = 200/200 (100%). Paired risk difference (guarded - baseline) = 0.02; preregistered exact McNemar two-sided p = 0.125 (primary confirmatory test). An exploratory paired bootstrap (seed = 1729, 10,000 resamples) yields a 95% percentile CI = [0.005, 0.04]. Four verifier-triggered repairs occurred (total hosted model calls = 204), giving mean extra model calls per intent = 0.02 and mean extra calls per avoided violation = 1.0. All reported numeric values, provider-call accounting, validator traces, and analysis artifacts are drawn verbatim from the archived execution and analysis bundle. We interpret the preregistered confirmatory result conservatively and discuss construct, internal, and external validity limits and operational implications.**

I. Introduction

Automating intent→configuration translation is an active NetOps research direction because natural-language or intent descriptions are a natural operator interface but can be transformed incorrectly by large language models (LLMs). Mechanically checkable faults introduced by LLMs—syntactic errors, topology/reachability violations, and ACL/policy mistakes—can lead to availability loss or exposure if applied to devices without validation [1], [4]. Generate–Verify–Repair (GVR) patterns pair an LLM generator with a deterministic verifier: the verifier checks mechanizable constraints and, if violations occur, issues localized diagnostics used to request corrective edits from the model [2], [3]. Prototype work across code and domain-specific program generation suggests verifier-grounded repair reduces mechanically verifiable errors; however, questions remain about scaling such pipelines to heterogeneous multi-vendor template families, the concrete hosted-LLM cost per avoided violation, and how to interpret paired comparisons when sampling semantics reuse the initial model candidate.

Research question and contribution. Our preregistered research question is: Does a verifier-grounded generate–verify-repair loop significantly reduce operationally detectable policy-violation errors produced by a hosted LLM when generating network configurations for operator intents across heterogeneous multi-vendor template families and topology patterns in a Mininet-based emulation? Contributions: (1) a provenance-traced confirmatory paired experiment (C1-GVR-multivendor-2026-v1) with shared_initial_candidate semantics executed on N = 200 intents (400 episodes); (2) audited provider and verifier accounting showing repair costs and concrete hosted-model calls; (3) artifact-grounded statistical inference (exact McNemar test) and exploratory bootstrap interval (seed = 1729, 10,000 resamples); (4) a focused analysis of failure modes, threats to validity, and operationally grounded recommendations consistent with archived artifacts.

II. Related Work

Recent surveys and reviews document rapid growth in LLM-for-NetOps and AIOps research while emphasizing recurring evaluation gaps—benchmark provenance, verifier integration, and limited multi-vendor emulation evidence [5], [6]. Empirical prototypes and benchmarks show that LLMs can draft device configurations or configuration-style artifacts, but they frequently produce mechanically checkable faults (syntax, missing or misordered commands, and reachability or policy violations) when evaluated on configuration translation tasks; these failure modes have been observed in domain-specific studies and NetOps benchmarks [1], [4].

A separate strand of work proposes and demonstrates verifier-grounded closed-loop architectures (generate→verify→repair) in coding and domain-specific program generation settings [2], [3]. These proposals and single-case or small-scale empirical demonstrations indicate that when (i) a deterministic verifier can express the relevant mechanizable constraints, and (ii) the verifier provides localized, actionable diagnostics, a bounded repair policy can reduce the incidence of mechanically verifiable errors. However, the cited records are prototype-scale: effectiveness depends on verifier expressivity, the class of detectable violations, and close coupling between verifier diagnostics and repair prompts; they do not by themselves establish broad production-scale safety or generality across heterogeneous vendor templates and topologies [2], [3].

Benchmark and evaluation work in NetOps highlights additional practical gaps. NetConfEval and related evaluations demonstrate feasibility of measuring LLM capability on configuration tasks, but these efforts commonly lack provenance-traced verifier pipelines and do not reliably quantify multi-vendor template heterogeneity or cost metrics tied to verifier-triggered repairs [4]. The surveys cited above call for richer provenance, clearer operationally meaningful metrics, and higher-fidelity emulation in order to make comparative claims about reliability and to expose residual semantic failures that deterministic verifiers may miss [5], [6].

Taken together, the verified literature supports three tempered conclusions that motivate this study: (1) LLMs can produce useful draft configurations but commonly make mechanically checkable errors; (2) verifier-grounded repair is a recurring mitigation pattern with promising single-case evidence but limited large-scale cross-vendor validation; and (3) more provenance-traced, operationally framed experiments are required to quantify repair cost tradeoffs and the residual semantic errors outside verifier coverage. Our preregistered paired design directly targets these gaps by providing provenance-traced accounting, an explicit shared_initial_candidate execution semantics to isolate verifier/repair effects, and per-intent accounting of repair calls and final validator outcomes.

III. Methodology

Preregistration and primary estimand. The study was preregistered as C1-GVR-multivendor-2026-v1 prior to any model calls (preregistration SHA-256: 9e0fc992665835975b01289f7841926ef6abf3584252c5d0 459ba0ecf166bb50). The primary estimand is the paired_success_rate_difference_guarded_minus_baseline: the per-intent difference in the probability that an intent yields zero operationally detectable violations under the guarded pipeline versus baseline. The preregistered confirmatory test is an exact two-sided McNemar test at $\alpha = 0.05$. Predeclared exploratory summaries include a paired bootstrap percentile CI (bootstrap_seed = 1729; resamples = 10,000) and descriptive hosted-model cost metrics (mean extra calls per intent and mean extra calls per avoided violation).

Sampling design, deterministic task generation, and strata. Task indices (N = 200 unique intents) were selected deterministically by the adapter before any model interactions. The sampling plan was stratified across three synthetic vendor-template families and four topology patterns (12 strata) to exercise template heterogeneity and workflow policy variety. Task payloads were generated by deterministic_netops_task_generator_v5 and include explicit per-task constraints (allowed_touched_fields, workflow_policies, required_sequence) and a canonical expected_final_state used by the deterministic verifier. The deterministic task generation and the fixed task_index_profile ensure that no post-hoc selection or data-dependent task filtering affected the confirmatory analysis.

Shared_initial_candidate semantics and paired condition definitions. Execution used shared_initial_candidate semantics as preregistered and archived: for each intent exactly one initial sampled generation was produced (the shared initial candidate) and that same initial text was presented to both baseline and guarded scoring/validation pipelines. Under these semantics: (a) baseline outcome equals the deterministic validator's verdict (validation_before) on the shared initial candidate; (b) guarded outcome equals the final guarded-pipeline verdict after deterministic validation and any permitted repair(s). The preregistration explicitly declared that independent constrained-prompt arms were not executed; therefore any paired discordance can only arise from deterministic validator behavior and from the allowed repair call(s), not from differing first-pass samples or prompt variants.

Model configuration, sampling notation, and audited provider budget. The declared model (as recorded in the archived model_configuration.json) is gpt-5-mini (provider recorded as openai_responses). The archived model_configuration.json records temperature = null/unset; null/unset is not evidence of deterministic hosted-provider sampling and does not imply temperature = 0. The adapter enforced the preregistered model-call budget: initial_generation_calls_per_task = 1, maximum_repair_calls_per_task = 1, maximum_model_calls = 400, planned_episode_count = 400. Provider audit fields in the execution manifest (hosted_model_call_event_count, stage_counts, condition_counts, cache_hit/miss counts, and cross_condition_cache_key_reuse_observed) were used to reconcile actual model calls and to verify that cross-condition cache reuse did not occur.

Deterministic verifier architecture and scoring rule. The deterministic_netops_validator_v5 is a rule-based verifier combining multiple mechanizable checks applied against the Mininet emulated state and the per-task payload: (1) syntactic parsing and command pattern matching to detect malformed or unrecognized commands; (2) required-command sequencing and workflow policy checks that ensure the produced sequence can satisfy required_sequence constraints and group/ordering invariants; (3) reachability/topology checks that simulate or reason over the emulated final_state to detect unsafe shutdowns or simultaneous outages; and (4) ACL/policy checks encoded from task-specific policy constraints. For each evaluated candidate the verifier returns a structured validator_trace containing normalized_configuration, parsed_command_count, required_command_count, observed_touched_field_count, violation_codes (if any), violation_count, and boolean valid. The scoring rule full_intent_constraint_validation yields score = 1 (success) only when violation_count == 0 and all required mechanizable constraints are satisfied.

Repair policy and repair-prompt formation. The guarded pipeline permits up to one verifier-triggered repair call per intent and only issues a repair call when the verifier reports one or more violations on the shared initial candidate. Repair prompts are deterministic templates that include: the verifier's localized diagnostics (violation_codes and short natural-language diagnostics), the normalized shared initial_configuration, the task's required_sequence and workflow constraints, and a request for a corrected configuration sequence that preserves unspecified state per the task payload. Repair prompts do not alter the initial prompt template family; they request corrective edits against the shared_initial_candidate. When a repair response is returned, the verifier is re-applied; if validation_after reports no violations the guarded outcome is success. Repair Provider traces and repair_prompt SHA references are archived for each repaired pair.

Provider-call accounting and stage_counts interpretation. All provider call accounting reported in the execution manifest and deterministic_reconciliation was reconciled deterministically. The audited hosted_model_call_event_count = 204 decomposes as 200 shared_initial_generation events (one per intent) plus 4 repair calls (repair stage). The execution manifest stage_counts field records {"shared_initial_generation": 200, "repair": 4}. The condition_counts mapping in the manifest (shown as {"shared": 200, "guarded": 4}) corresponds to these stage counts under the shared_initial_candidate semantics: the label "shared" enumerates the initial sampled generations produced for reuse, while the label "guarded" in the manifest reflects the number of repair-stage calls actually invoked by the verifier. The manifest also records cache_hit_count = 0 and cross_condition_cache_key_reuse_observed = false, supporting that the same shared initial candidate was deterministically reused by both baseline and guarded conditions rather than created by cross-condition cache reuse.

Scoring decisions, exclusions, and missingness treatment. Scoring decisions followed the preregistered contract: ambiguous verifier outputs (those flagged by deterministic ambiguity rules) were conservatively treated as violations in the primary analysis. Exclusion rules predeclared duplicate tasks (detected by deterministic prompt+context hashing), unrecoverable container/template substitution failures, and infrastructure integrity failures; excluded pairs would be reported separately and included in sensitivity analyses. The primary analysis used the 'complete_pair_primary' treatment: only complete pairs (both episodes scored) were included. The preregistered sensitivity analysis ('complete_pair_primary_with_failure_as_zero_sensitivity') treats unrecoverable or aborted pairs conservatively as failures; no such exclusions occurred in the archived run (complete_pair_count = 200; failed_episode_count = 0).

Analysis procedures and bootstrap caution. Primary inferential analysis is the exact two-sided McNemar test on the paired binary success indicators (baseline vs guarded). Secondary exploratory summaries include an empirical paired bootstrap percentile CI for the paired risk difference (bootstrap_seed = 1729, resamples = 10,000) and descriptive statistics for model-call costs (mean extra calls per intent, mean extra calls per avoided violation). Because paired bootstrap percentile intervals for binary paired differences can be unstable when the total number of discordant pairs (n10 + n01) is small, we treat the bootstrap CI as exploratory magnitude information only and emphasize the exact McNemar p-value as the preregistered confirmatory outcome. Analysis code, seeds, and resampling parameters are archived and referenced in analysis_log.jsonl.

IV. Results

Execution and audited provider accounting. The preregistered run completed as planned: planned_episode_count = 400; completed_episode_count = 400; complete_pair_count = 200; failed_episode_count = 0. Deterministic provider accounting (execution manifest) reports hosted_model_call_event_count = 204 decomposed into 200 shared_initial_generation events and 4 repair calls; cache_hit_count = 0; cache_miss_count = 204; stage_counts = {shared_initial_generation: 200, repair: 4}. These audit counts are reported verbatim from the archived execution manifest.

Paired outcomes and confirmatory test. The paired 2×2 contingency (analysis/paired_contingency_table.csv) is: n11 = 196 (both succeed), n10 = 4 (guarded succeed, baseline fail), n01 = 0 (guarded fail, baseline succeed), n00 = 0 (both fail). Baseline success rate = 196/200 = 0.98; guarded success rate = 200/200 = 1.0; paired risk difference (guarded - baseline) = 0.02. The preregistered exact McNemar two-sided test yields p = 0.125 (primary confirmatory result; do not reject at α = 0.05).

Exploratory interval and bootstrap caution. The archived exploratory paired bootstrap (bootstrap_seed = 1729; resamples = 10,000) produced a 95% percentile CI for the paired risk difference = [0.005, 0.04]. We report this interval as exploratory and caution that bootstrap percentile intervals can be unstable when discordant pair counts (n10 + n01) are small; the exact McNemar test remains the formal preregistered inference. The bootstrap seed and resampling count are archived (analysis_log.jsonl).

Repair events and cost accounting. Four verifier-triggered repairs were invoked across 200 intents. Total hosted model calls = 204 (200 shared initial + 4 repairs), giving mean hosted model calls per intent = 1.02 and mean extra calls per intent due to repairs = 0.02. Each repair corrected one baseline failure in the archived contingency; thus mean extra calls per avoided violation = 1.0. These arithmetic derivations follow directly from the audited provider_call counts in deterministic_reconciliation and the execution manifest.

Representative artifacts and substantiation. Representative scoring artifacts (execution/scoring/task-000004-*, task-000005-*, task-000006-*) show identical shared_initial_candidate content and identical shared_initial responses reused by both baseline and guarded conditions. For the four discordant pairs, validator_trace.validation_before recorded detected violations and validation_after recorded that a repair was applied and the final configuration passed full_intent_constraint_validation. The raw shared_initial response files, validator traces, and repair decisions are archived and cited in the execution manifest; they substantiate the contingency counts and repair accounting above.

Power and interpretation. The preregistered power analysis targeted a 10 percentage point absolute reduction with N $\approx$ 157 (McNemar approximation); N = 200 was chosen to permit stratified descriptions. The observed baseline failure prevalence (2%) was lower than some planning scenarios, reducing power to detect small absolute improvements and producing the small discordant count (4) that underlies the non-significant McNemar result. We therefore interpret the p = 0.125 as the definitive confirmatory outcome under the preregistered design and treat the bootstrap CI as supplementary magnitude information.

V. Discussion

Causal interpretation under shared_initial_candidate semantics. Because execution used shared_initial_candidate semantics (one sampled initial generation per intent reused by baseline and guarded), paired improvements arise solely from deterministic validator/repair actions. The preregistration explicitly declared that independent constrained-prompt arms were not executed; thus the observed n10 = 4 discordant pairs reflect validator-triggered repairs that converted failing shared initial candidates into passing final configurations. Any statements attributing improvements to prompt-engineering or differing first-pass samples would be incorrect for this run.

Construct validity and verifier coverage. The deterministic_netops_validator_v5 covers syntactic correctness, required command sequencing, reachability/topology checks against the emulated state, and encoded ACL/policy checks derived from per-task payload constraints. This verifier can only detect violations expressible in its rules; higher-order semantic misinterpretations not captured by those encodings remain undetectable and are identified in our limitations. Increasing verifier expressivity (formal policy ontologies, broader semantic checks) is a direct next step to reduce residual false negatives.

Internal validity and nondeterminism. The archived model_configuration.json records temperature = null/unset; we reiterate that null/unset does not imply temperature = 0 or deterministic provider sampling. The execution manifest's provider audit (hosted_model_call_event_count = 204, stage_counts, and condition_counts) was reconciled deterministically (all deterministic consistency checks passed) and confirms that cross-condition cache reuse did not occur; this audit supports the claim that baseline outcomes are the validator verdict on the identical shared initial generation used by guarded episodes.

Operational implications (bounded). Within the constrained Mininet emulation, synthetic templates, and validator rule set, a small repair budget (four repairs) removed the mechanically detectable baseline violations at very low hosted-model cost. This artifact-grounded observation suggests that when a deterministic verifier exists and provides localized actionable diagnostics, a bounded repair policy can remove template-driven mechanically checkable faults in similar emulated settings. We do not claim production readiness, multi-model generality, nor coverage for semantic errors outside the verifier.

Threats to validity and recommendations. Internal validity—shared_initial_candidate semantics were predeclared and executed; this prevents attributing gains to independent re-sampling. Construct validity—the verifier cannot detect all semantic errors; extend verifier coverage and measure false negatives. External validity—single declared model (gpt-5-mini) and synthetic templates in Mininet limit generalization; evaluate additional hosted models and hardware testbeds. Conclusion validity—small discordant counts reduce power; future preregistered work could target higher baseline failure prevalence scenarios or increase N. Based on archived evidence we recommend: (1) expanding deterministic verifier expressivity to formalize more semantic/policy ontologies; (2) performing paired designs that also execute independent multi-sample initial generations to separate prompt/sampling effects from repair effects; and (3) repeating provenance-traced evaluations across multiple hosted models and template families.

VI. Limitations

- Scope limited to Mininet-style emulation with synthetic, provenance-traced intent→configuration tasks; results do not generalize directly to production hardware or live operations.
- Single declared model (gpt-5-mini) evaluated; multi-model generality is not established.
- Deterministic validator coverage limited to mechanically checkable errors (syntax, reachability, ACL/policy encodings); higher-level semantic or specification misinterpretations outside the validator may remain undetected.
- Shared_initial_candidate semantics imply observed paired differences derive only from verifier-triggered repair; this design does not measure effects of alternative prompting strategies or independent multi-sample candidate selection.
- Observed confirmatory effect (2 percentage points) did not reach statistical significance at α = 0.05 for N = 200 (exact McNemar p = 0.125); low baseline failure prevalence (2%) reduced power to detect small absolute improvements under some preregistered scenarios.

VII. Conclusion

We preregistered and executed a provenance-traced paired confirmatory study (C1-GVR-multivendor-2026-v1) using shared_initial_candidate semantics to measure whether a deterministic verifier plus bounded repair reduces mechanically detectable policy violations for intent→configuration tasks in a Mininet emulation. The preregistered exact McNemar two-sided test did not reject at α = 0.05 (p = 0.125). Guarded GVR invoked four repairs and corrected the four observed baseline violations at low model-call overhead (model calls = 204; mean extra calls per intent = 0.02), yielding a paired risk difference estimate of 0.02 and an exploratory paired bootstrap CI = [0.005, 0.04] (bootstrap_seed = 1729, resamples = 10,000). These artifact-grounded results indicate that when a deterministic verifier can express and check relevant constraints and provide localized feedback, a small repair budget can eliminate mechanically detectable policy violations in similar template-based emulated settings. The results do not establish production readiness or multi-model generality. All reported numbers, provider accounting, validator traces, and analysis artifacts are drawn verbatim from the archived execution and analysis bundles referenced in the preregistration and execution manifests.

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Disclosure Statement

Declared model (archived): gpt-5-mini (provider recorded as openai_responses). Adapter family: hosted_netops_gvr_v1. Deterministic validator: deterministic_netops_validator_v5. Task generator: deterministic_netops_task_generator_v5. Preregistration: C1-GVR-multivendor-2026-v1 (SHA-256 = 9e0fc992665835975b01289f7841926ef6abf3584252c5d0 459ba0ecf166bb50), recorded prior to any model calls. Initial master prompt archived at provenance/master_prompt.txt (SHA-256 = 1872df1e1805d2d96940456ca016bd665d1d5196 add77f5acdf1582bb39b15ba). Human role: a human Research Director selected the topic family and constrained the initial master prompt prior to the autonomy lock; after the lock the AI autonomously performed literature verification, preregistration, experiment execution, analysis, peer review, revision, and manuscript drafting; no human manually edited technical manuscript content after the autonomy lock. Sampling semantics: execution used shared_initial_candidate (exactly one sampled initial generation per intent was produced and reused by both baseline and guarded); archived model configuration records temperature = null/unset (null/unset is not evidence of deterministic provider sampling and does not imply temperature = 0). Provider accounting (audited): hosted_model_call_event_count = 204 decomposed as 200 shared initial generations + 4 repair calls. Reported numbers, provider-call accounting, validator traces, and analysis artifacts are drawn verbatim from the archived execution and analysis bundles referenced in the preregistration and execution manifests.

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